

Solving Non-Homogeneous Linear Recurrence Relations

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1. General Approach

Given a recurrence relation of the form:

$$a_{n+k} + c_{k-1}a_{n+k-1} + \cdots + c_0a_n = f(n)$$

we solve it in two parts:

1. The **homogeneous solution** $a_n^{(h)}$ by solving the characteristic equation.
2. The **particular solution** $a_n^{(p)}$ depending on the form of the forcing function $f(n)$.

2. Example 1: Polynomial Forcing

$$a_{n+2} = 4a_{n+1} - 4a_n + n^2.$$

Step 1: Solve the homogeneous part

$$a_{n+2} - 4a_{n+1} + 4a_n = 0.$$

Characteristic equation: $(r - 2)^2 = 0 \Rightarrow$ root $r = 2$ (multiplicity 2).

$$a_n^{(h)} = (C_1 + C_2n)2^n.$$

Step 2: Particular solution guess Since $f(n) = n^2$ is a polynomial of degree 2 and does not conflict with the root $r = 2$ (i.e. not of form $A2^n$), assume:

$$a_n^{(p)} = An^2 + Bn + C.$$

Substitute into the recurrence and solve for A, B, C . Result:

$$a_n^{(p)} = n^2 + 4n + 8.$$

Step 3: General solution

$$a_n = (C_1 + C_2n)2^n + n^2 + 4n + 8.$$

For $a_0 = 1, a_1 = 2$:

$$a_n = (-7 + \frac{3}{2}n)2^n + n^2 + 4n + 8.$$

3. When $f(n) = r^n$

If the non-homogeneous part is $f(n) = r^n$, assume a particular solution of the form $a_n^{(p)} = Ar^n$ unless r is a root of the characteristic equation.

- If r is **not a root**, use Ar^n .
- If r is a **simple root**, multiply by n : Anr^n .
- If r is a **double root**, multiply by n^2 : An^2r^n .

4. When $f(n) = nr^n$

If $f(n) = nr^n$ and r is a root of multiplicity s , then multiply by n^s :

$$a_n^{(p)} = n^s(An + B)r^n.$$

Example:

$$a_{n+2} - 4a_{n+1} + 3a_n = n3^n.$$

Characteristic equation: $(r - 3)(r - 1) = 0$. Roots: 3, 1.

Since forcing term includes 3^n and 3 is a root (multiplicity 1), try:

$$a_n^{(p)} = n(An + B)3^n = (An^2 + Bn)3^n.$$

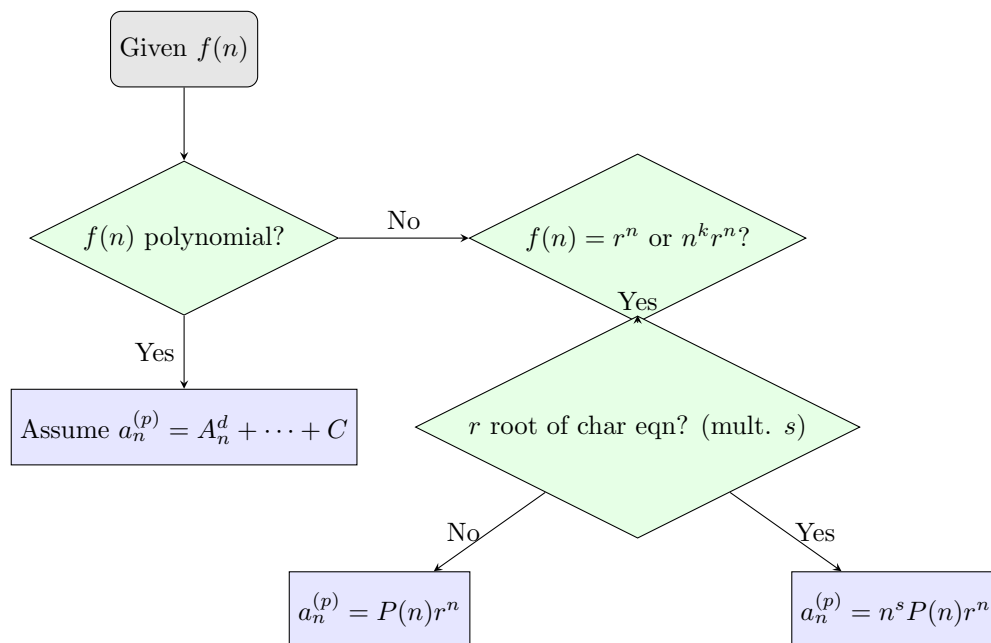
Substituting and solving yields:

$$a_n^{(p)} = (\frac{1}{12}n^2 - \frac{1}{3}n)3^n.$$

General solution:

$$a_n = C_13^n + C_2 + (\frac{1}{12}n^2 - \frac{1}{3}n)3^n.$$

5. Decision Flowchart for Choosing $a_n^{(p)}$



Recurrence Relation Problems

Problem 1

Solve the recurrence:

$$a_n = 4a_{n-1} - 4a_{n-2} + n^2 \cdot 2^n, \quad a_1 = 1, a_2 = 4.$$

Solution:

1. Homogeneous part.

$$r^2 - 4r + 4 = 0 \quad \Rightarrow \quad (r - 2)^2 = 0.$$

Thus, the homogeneous solution is

$$a_n^{(h)} = (C_1 + C_2 n) 2^n.$$

2. Particular part. Since the non-homogeneous term is $n^2 \cdot 2^n$ and $r = 2$ is a double root, we try

$$a_n^{(p)} = (\alpha n^2 + \beta n + \gamma) n^2 2^n.$$

The $(\alpha n^2 + \beta n + \gamma) 2^n$ part is from the original equation and the n^2 is becoz of repeated roots (2 being a repeated root of the homogenous part) Substituting into the recurrence and equating coefficients yields

$$\alpha = \frac{1}{12}, \quad \beta = \frac{1}{3}, \quad \gamma = \frac{5}{12}.$$

Hence

$$a_n^{(p)} = 2^n \left(\frac{n^4 + 4n^3 + 5n^2}{12} \right).$$

3. General solution.

$$a_n = 2^n \left(C_1 + C_2 n + \frac{n^4 + 4n^3 + 5n^2}{12} \right).$$

4. Apply initial conditions.

$$\begin{cases} a_1 = 1 = 2(C_1 + C_2 + \frac{10}{12}), \\ a_2 = 4 = 4(C_1 + 2C_2 + \frac{68}{12}). \end{cases}$$

Solving gives $C_1 = 4$, $C_2 = -\frac{13}{3}$.

Final closed form:

$$a_n = 2^n \left(4 - \frac{13}{3}n + \frac{n^2(n^2 + 4n + 5)}{12} \right)$$

or equivalently,

$$a_n = 2^n \cdot \frac{n^4 + 4n^3 + 5n^2 - 52n + 48}{12}.$$

Problem 2

Solve the recurrence:

$$a_n = 4a_{n-1} - 3a_{n-2} + n^2 \cdot 3^n, \quad a_1 = 1, \quad a_2 = 4.$$

Solution:

1. Homogeneous part.

$$r^2 - 4r + 3 = 0 \quad \Rightarrow \quad r = 1, 3.$$

Thus,

$$a_n^{(h)} = C_1 + C_2 \cdot 3^n.$$

2. Particular part. The non-homogeneous term $n^2 \cdot 3^n$ involves a simple root $r = 3$, so we try

$$a_n^{(p)} = 3^n(\alpha n^3 + \beta n^2 + \gamma n).$$

Substituting and matching coefficients gives

$$\alpha = \frac{1}{2}, \quad \beta = 0, \quad \gamma = 1.$$

Hence

$$a_n^{(p)} = 3^n \left(\frac{1}{2}n^3 + n \right).$$

3. General solution.

$$a_n = C_1 + 3^n \left(C_2 + \frac{1}{2}n^3 + n \right).$$

4. Apply initial conditions.

$$\begin{cases} a_1 = 1 = C_1 + 3(C_2 + \frac{1}{2} + 1), \\ a_2 = 4 = C_1 + 9(C_2 + 4 + 2). \end{cases}$$

Solving gives $C_1 = \frac{79}{4}$, $C_2 = -\frac{31}{4}$.

Final closed form:

$$a_n = \frac{79}{4} + 3^n \left(-\frac{31}{4} + \frac{1}{2}n^3 + n \right)$$

or equivalently,

$$a_n = \frac{79 + 3^n(2n^3 + 4n - 31)}{4}.$$